

PROJECT BASED EXPERIMENTAL LEARNING

COURSE- DATABASE MANAGEMENT SYSTEM (NoSQL / MongoDB)

PROJECT REPORT

TOPIC- MONGODB STUDENT DATABASE OPERATIONS



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Abstract

This report presents a MongoDB-based Student Database Management project that demonstrates the core Create, Read, Update and Delete (CRUD) operations of a NoSQL document database. A "students" collection is used to store student records containing roll number, name, age, department and marks. The project implements single and bulk insertion, conditional updates, single and bulk deletion, and a wide range of query operations using MongoDB comparison and logical operators such as \$gt, \$lt, \$eq, \$gte, \$ne, \$and, \$or and \$not. The implementation demonstrates how MongoDB can be used to efficiently manage, filter and retrieve structured student data.

Introduction

MongoDB is a popular NoSQL, document-oriented database that stores data as flexible, JSON-like documents rather than in rigid rows and columns. This makes it well suited for applications where the data structure may evolve over time, such as academic record-keeping systems. This project focuses on building and querying a "students" collection that stores academic details of 51 students, and demonstrates the standard database operations - insertion, updation, deletion and querying - that form the backbone of any real-world database-driven application.

Methodology

The project was carried out step by step inside the MongoDB shell (mongosh), with every operation executed against a single collection named "students" in a database created for this purpose. The complete methodology, along with the exact code executed and the output produced by MongoDB at each step, is given below.

1. Insert One

Inserts a single student document into the students collection.

Code:

```
db.students.insertOne({
  rollNo: 101,
  name: "Rahul",
  age: 20,
  department: "CSE",
  marks: 85
})
```

2. Insert Many

Inserts multiple student documents (roll numbers 102 to 151) into the collection at the same time.

Code:

```
db.students.insertMany([
{ rollNo: 102, name: "Priya", age: 19, department: "IT", marks: 78 },
{ rollNo: 103, name: "Amit", age: 21, department: "CSE", marks: 92 },
{ rollNo: 104, name: "Sneha", age: 20, department: "ECE", marks: 81 },
{ rollNo: 105, name: "Arjun", age: 22, department: "ME", marks: 74 },
{ rollNo: 106, name: "Neha", age: 19, department: "CSE", marks: 88 },
{ rollNo: 107, name: "Rohit", age: 20, department: "IT", marks: 69 },
{ rollNo: 108, name: "Anjali", age: 21, department: "ECE", marks: 95 },
{ rollNo: 109, name: "Vikas", age: 20, department: "CSE", marks: 83 },
{ rollNo: 110, name: "Pooja", age: 19, department: "IT", marks: 76 },
{ rollNo: 111, name: "Karan", age: 22, department: "ME", marks: 72 },
{ rollNo: 112, name: "Riya", age: 20, department: "CSE", marks: 89 },
{ rollNo: 113, name: "Sahil", age: 21, department: "ECE", marks: 67 },
{ rollNo: 114, name: "Kavya", age: 19, department: "IT", marks: 91 },
{ rollNo: 115, name: "Nikhil", age: 20, department: "CSE", marks: 79 },
{ rollNo: 116, name: "Simran", age: 21, department: "ECE", marks: 84 },
{ rollNo: 117, name: "Aditya", age: 22, department: "ME", marks: 73 },
{ rollNo: 118, name: "Muskan", age: 20, department: "CSE", marks: 96 },
{ rollNo: 119, name: "Varun", age: 19, department: "IT", marks: 68 },
{ rollNo: 120, name: "Shreya", age: 21, department: "ECE", marks: 87 },
{ rollNo: 121, name: "Manish", age: 20, department: "CSE", marks: 75 },
{ rollNo: 122, name: "Nisha", age: 19, department: "IT", marks: 90 },
{ rollNo: 123, name: "Akash", age: 22, department: "ME", marks: 71 },
{ rollNo: 124, name: "Divya", age: 20, department: "CSE", marks: 86 },
{ rollNo: 125, name: "Harsh", age: 21, department: "ECE", marks: 80 },
{ rollNo: 126, name: "Isha", age: 19, department: "IT", marks: 94 },
{ rollNo: 127, name: "Yash", age: 20, department: "CSE", marks: 77 },
{ rollNo: 128, name: "Komal", age: 21, department: "ECE", marks: 82 },
{ rollNo: 129, name: "Deepak", age: 22, department: "ME", marks: 65 },
{ rollNo: 130, name: "Tanya", age: 20, department: "CSE", marks: 93 },
{ rollNo: 131, name: "Mohit", age: 19, department: "IT", marks: 70 },
{ rollNo: 132, name: "Sakshi", age: 21, department: "ECE", marks: 89 },
{ rollNo: 133, name: "Rajat", age: 20, department: "CSE", marks: 81 },
{ rollNo: 134, name: "Payal", age: 19, department: "IT", marks: 85 },
{ rollNo: 135, name: "Abhishek", age: 22, department: "ME", marks: 78 },
{ rollNo: 136, name: "Aarti", age: 20, department: "CSE", marks: 97 },
{ rollNo: 137, name: "Gaurav", age: 21, department: "ECE", marks: 74 },
{ rollNo: 138, name: "Preeti", age: 19, department: "IT", marks: 88 },
{ rollNo: 139, name: "Suresh", age: 20, department: "CSE", marks: 66 },
{ rollNo: 140, name: "Megha", age: 21, department: "ECE", marks: 92 },
{ rollNo: 141, name: "Ravi", age: 22, department: "ME", marks: 69 },
{ rollNo: 142, name: "Shivani", age: 20, department: "CSE", marks: 84 },
{ rollNo: 143, name: "Ankit", age: 19, department: "IT", marks: 79 },
{ rollNo: 144, name: "Pallavi", age: 21, department: "ECE", marks: 91 },
{ rollNo: 145, name: "Rohan", age: 20, department: "CSE", marks: 87 },
{ rollNo: 146, name: "Monika", age: 19, department: "IT", marks: 73 },
{ rollNo: 147, name: "Varsha", age: 21, department: "ECE", marks: 86 },
{ rollNo: 148, name: "Pankaj", age: 22, department: "ME", marks: 68 },
{ rollNo: 149, name: "Ritika", age: 20, department: "CSE", marks: 90 },
{ rollNo: 150, name: "Naveen", age: 21, department: "IT", marks: 76 },
{ rollNo: 151, name: "Sonia", age: 19, department: "ECE", marks: 83 }
```

```
])
```

3. Update One

Updates the first document that matches the given condition - here, the marks of roll number 101 are updated from 85 to 90.

Code:

```
db.students.updateOne(  
  { rollNo: 101 },  
  { $set: { marks: 90 } }  
)
```

Output:

```
{  
  acknowledged: true,  
  matchedCount: 1,  
  modifiedCount: 1  
}
```

4. Update Many

Updates all documents that match the given condition - here, every student in the CSE department is given a new status field set to "Active".

Code:

```
db.students.updateMany(  
  { department: "CSE" },  
  { $set: { status: "Active" } }  
)
```

Output:

```
{  
  acknowledged: true,  
  matchedCount: 16,  
  modifiedCount: 16  
}
```

5. Delete One

Deletes the first document that matches the given condition - here, the record of roll number 151 is removed.

Code:

```
db.students.deleteOne({ rollNo: 151 })
```

Output:

```
{  
  acknowledged: true,  
  deletedCount: 1  
}
```

6. Delete Many

Deletes all documents that match the given condition - here, every student with marks less than 40 is removed.

Code:

```
db.students.deleteMany({ marks: { $lt: 40 } })
```

Output:

```
{
  acknowledged: true,
  deletedCount: 0
}
```

7. Query Operations

A series of find() queries were run using MongoDB's comparison and logical operators to filter and retrieve student records.

Marks greater than 80:

Code:

```
db.students.find({ marks: { $gt: 80 } })
```

Output:

```
103 Amit    21 CSE 92
104 Sneha   20 ECE 81
106 Neha    19 CSE 88
108 Anjali  21 ECE 95
109 Vikas   20 CSE 83
112 Riya    20 CSE 89
114 Kavya   19 IT  91
116 Simran  21 ECE 84
118 Muskan  20 CSE 96
120 Shreya  21 ECE 87
122 Nisha   19 IT  90
124 Divya   20 CSE 86
126 Isha    19 IT  94
128 Komal   21 ECE 82
130 Tanya   20 CSE 93
132 Sakshi  21 ECE 89
133 Rajat   20 CSE 81
134 Payal   19 IT  85
136 Aarti   20 CSE 97
138 Preeti  19 IT  88
140 Megha   21 ECE 92
142 Shivani 20 CSE 84
144 Pallavi 21 ECE 91
145 Rohan   20 CSE 87
147 Varsha  21 ECE 86
149 Ritika  20 CSE 90
```

Marks less than 80:

Code:

```
db.students.find({ marks: { $lt: 80 } })
```

Output:

```
102 Priya   19 IT  78
105 Arjun   22 ME  74
107 Rohit   20 IT  69
```

110	Pooja	19	IT	76
111	Karan	22	ME	72
113	Sahil	21	ECE	67
115	Nikhil	20	CSE	79
117	Aditya	22	ME	73
119	Varun	19	IT	68
121	Manish	20	CSE	75
123	Akash	22	ME	71
127	Yash	20	CSE	77
129	Deepak	22	ME	65
131	Mohit	19	IT	70
135	Abhishek	22	ME	78
137	Gaurav	21	ECE	74
139	Suresh	20	CSE	66
141	Ravi	22	ME	69
143	Ankit	19	IT	79
146	Monika	19	IT	73
148	Pankaj	22	ME	68

Marks equal to 85:

Code:

```
db.students.find({ marks: { $eq: 85 } })
```

Output:

101	Rahul	20	CSE	85
134	Payal	19	IT	85

Marks greater than or equal to 80:

Code:

```
db.students.find({ marks: { $gte: 80 } })
```

Output:

101	Rahul	20	CSE	90
103	Amit	21	CSE	92
104	Sneha	20	ECE	81
106	Neha	19	CSE	88
108	Anjali	21	ECE	95
109	Vikas	20	CSE	83
112	Riya	20	CSE	89
114	Kavya	19	IT	91
116	Simran	21	ECE	84
118	Muskan	20	CSE	96
120	Shreya	21	ECE	87
122	Nisha	19	IT	90
124	Divya	20	CSE	86
125	Harsh	21	ECE	80
126	Isha	19	IT	94
128	Komal	21	ECE	82
130	Tanya	20	CSE	93
132	Sakshi	21	ECE	89
133	Rajat	20	CSE	81
134	Payal	19	IT	85
136	Aarti	20	CSE	97
138	Preeti	19	IT	88

```
140 Megha    21 ECE 92
142 Shivani  20 CSE 84
144 Pallavi  21 ECE 91
145 Rohan    20 CSE 87
147 Varsha   21 ECE 86
149 Ritika   20 CSE 90
```

Marks not equal to 85:

Code:

```
db.students.find({ marks: { $ne: 85 } })
```

Marks greater than 70 AND age less than 23:

Code:

```
db.students.find({
  $and: [
    { marks: { $gt: 70 } },
    { age: { $lt: 23 } }
  ]
})
```

Output:

```
101 Rahul    20 CSE 90
102 Priya    19 IT  78
103 Amit     21 CSE 92
104 Sneha    20 ECE 81
105 Arjun    22 ME  74
106 Neha     19 CSE 88
108 Anjali   21 ECE 95
109 Vikas    20 CSE 83
110 Pooja    19 IT  76
111 Karan    22 ME  72
112 Riya     20 CSE 89
114 Kavya    19 IT  91
115 Nikhil   20 CSE 79
116 Simran   21 ECE 84
117 Aditya   22 ME  73
118 Muskan   20 CSE 96
120 Shreya   21 ECE 87
121 Manish   20 CSE 75
122 Nisha    19 IT  90
123 Akash    22 ME  71
124 Divya    20 CSE 86
125 Harsh    21 ECE 80
126 Isha     19 IT  94
127 Yash     20 CSE 77
128 Komal    21 ECE 82
130 Tanya    20 CSE 93
132 Sakshi   21 ECE 89
133 Rajat    20 CSE 81
134 Payal    19 IT  85
135 Abhishek 22 ME  78
136 Aarti    20 CSE 97
137 Gaurav   21 ECE 74
138 Preeti   19 IT  88
```

```

140 Megha    21 ECE 92
142 Shivani  20 CSE 84
143 Ankit    19 IT  79
144 Pallavi  21 ECE 91
145 Rohan    20 CSE 87
146 Monika   19 IT  73
147 Varsha   21 ECE 86
149 Ritika   20 CSE 90
150 Naveen   21 IT  76

```

Department is CSE OR ECE:

Code:

```

db.students.find({
  $or: [
    { department: "CSE" },
    { department: "ECE" }
  ]
})

```

Output:

CSE and ECE department students are returned.

CSE:

```

101 Rahul      103 Amit      106 Neha      109 Vikas
112 Riya       115 Nikhil    118 Muskan    121 Manish
124 Divya      127 Yash      130 Tanya     133 Rajat
136 Aarti      139 Suresh    142 Shivani   145 Rohan
149 Ritika

```

ECE:

```

104 Sneha      108 Anjali    113 Sahil     116 Simran
120 Shreya     125 Harsh     128 Komal     132 Sakshi
137 Gaurav     140 Megha     144 Pallavi   147 Varsha
151 Sonia

```

Marks NOT greater than 80:

Code:

```

db.students.find({ marks: { $not: { $gt: 80 } } })

```

Output:

```

102 Priya      19 IT  78
105 Arjun      22 ME  74
107 Rohit      20 IT  69
110 Pooja      19 IT  76
111 Karan      22 ME  72
113 Sahil      21 ECE 67
115 Nikhil     20 CSE 79
117 Aditya     22 ME  73
119 Varun      19 IT  68
121 Manish     20 CSE 75
123 Akash      22 ME  71
125 Harsh      21 ECE 80
127 Yash       20 CSE 77
129 Deepak     22 ME  65

```

131	Mohit	19	IT	70
135	Abhishek	22	ME	78
137	Gaurav	21	ECE	74
139	Suresh	20	CSE	66
141	Ravi	22	ME	69
143	Ankit	19	IT	79
146	Monika	19	IT	73
148	Pankaj	22	ME	68
150	Naveen	21	IT	76

8. MongoDB Operator Summary

Operator	Meaning	Example
\$gt	Greater than	marks: { \$gt: 80 }
\$lt	Less than	marks: { \$lt: 80 }
\$eq	Equal to	marks: { \$eq: 85 }
\$gte	Greater than or equal to	marks: { \$gte: 80 }
\$ne	Not equal to	marks: { \$ne: 85 }
\$and	All conditions must be true	marks > 70 AND age < 23
\$or	At least one condition must be true	CSE OR ECE
\$not	Negates a condition	NOT (marks > 80)

Total dataset: 51 students (Roll No. 101-151).

Implementation Details

- The project is implemented using MongoDB and executed through the MongoDB Shell (mongosh).
- A single collection named "students" is used, with fields: rollNo, name, age, department and marks.
- insertOne() and insertMany() are used to populate the collection with 51 student records.
- updateOne() and updateMany() with the \$set operator are used to modify existing fields and add new ones (e.g. status).
- deleteOne() and deleteMany() are used to remove individual and bulk records based on a filter condition.
- find() is combined with comparison operators (\$gt, \$lt, \$eq, \$gte, \$ne) and logical operators (\$and, \$or, \$not) to retrieve specific subsets of student data.
- Each operation's acknowledgement object (acknowledged, matchedCount, modifiedCount, deletedCount) is used to verify that it executed as intended.

Results

All CRUD operations executed successfully on the students collection. The insertOne and insertMany operations populated the collection with 51 student records. The updateOne operation successfully changed one student's marks, while updateMany correctly modified all 16 CSE-department documents to add an "Active" status. The deleteOne operation removed one record and deleteMany found no records matching its filter, correctly reporting a deletedCount of 0. The query operations correctly filtered students by marks and department using comparison and logical operators, demonstrating that MongoDB's query engine returns accurate, condition-based results across a realistic-sized dataset.

Conclusion

This project highlights the practical use of MongoDB as a NoSQL database for managing structured academic data. By implementing insertion, update, deletion and a wide variety of query operations on a students collection, the project demonstrates a solid, hands-on understanding of MongoDB's Create, Read, Update and Delete (CRUD) capabilities as well as its comparison and logical query operators. The approach shows how a document database can be used to store and efficiently retrieve real-world tabular data such as student records.

Future Scope

- Use the MongoDB Aggregation Framework to compute department-wise averages, toppers and rankings.
- Add schema validation rules to enforce data types and required fields at the database level.
- Create indexes on frequently queried fields (e.g. marks, department) to improve performance.
- Build a front-end / web dashboard (e.g. using Node.js and Express) to interact with the database visually.
- Extend the collection with additional entities such as courses, attendance and results, and model relationships between them.

References

- MongoDB Official Documentation: <https://www.mongodb.com/docs/>
- MongoDB CRUD Operations: <https://www.mongodb.com/docs/manual/crud/>
- MongoDB Query Operators: <https://www.mongodb.com/docs/manual/reference/operator/query/>
- mongosh (MongoDB Shell) Documentation: <https://www.mongodb.com/docs/mongodb-shell/>

THANK YOU!